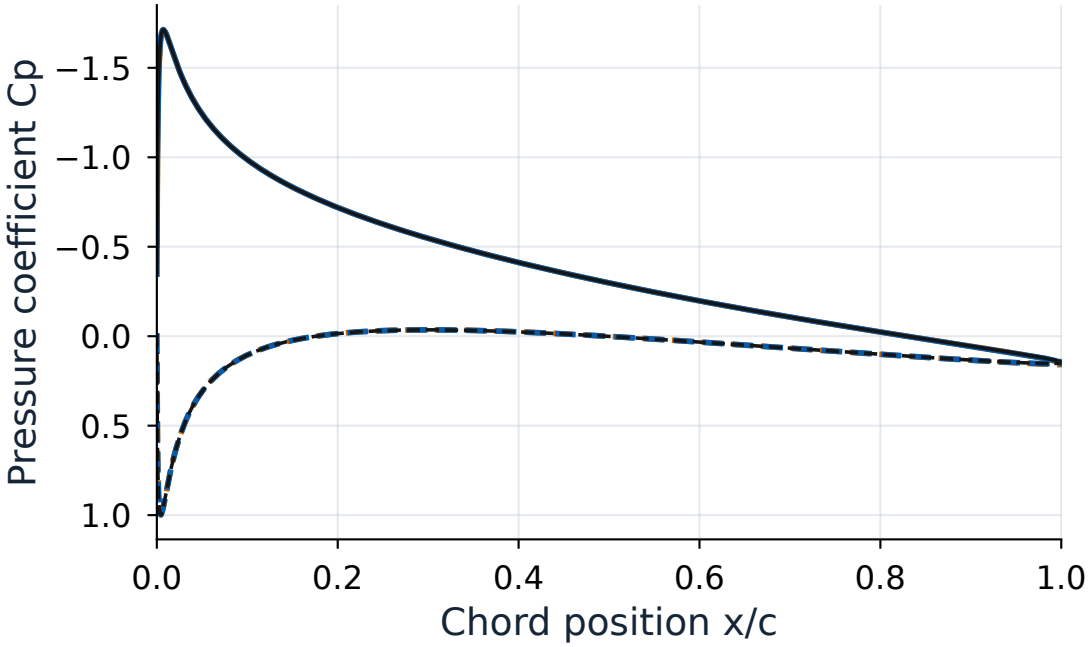
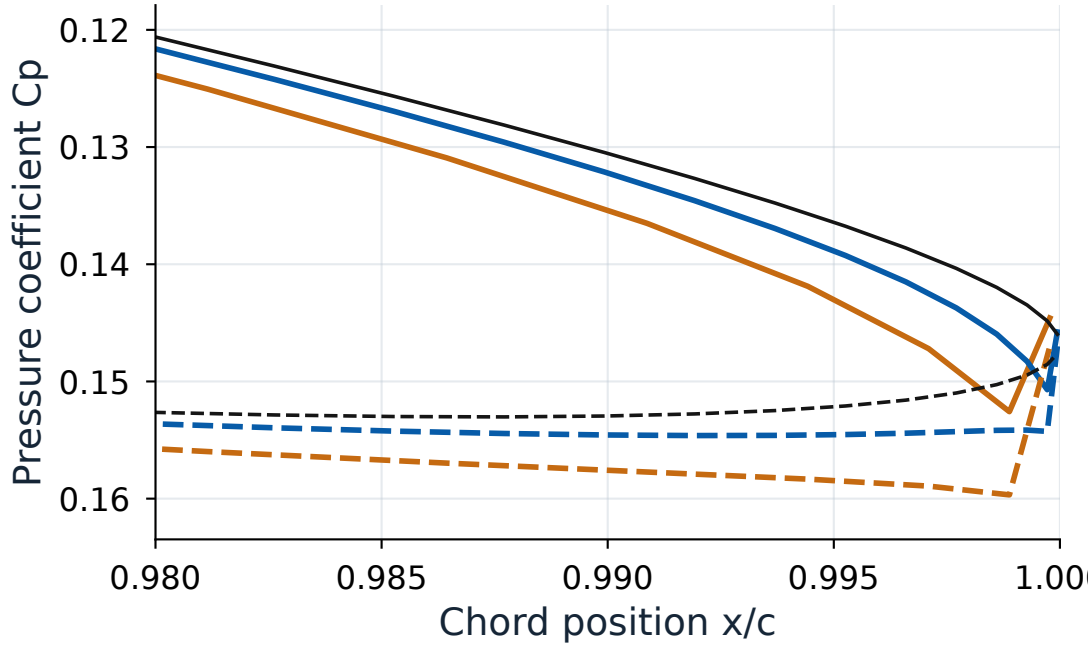
2D airfoil verification against an exact solution | angle of attack 4 degrees

Current solver: 160 panels
Current solver: 320 panels
Exact potential-flow solution
Solid: upper surface
Dashed: lower surface

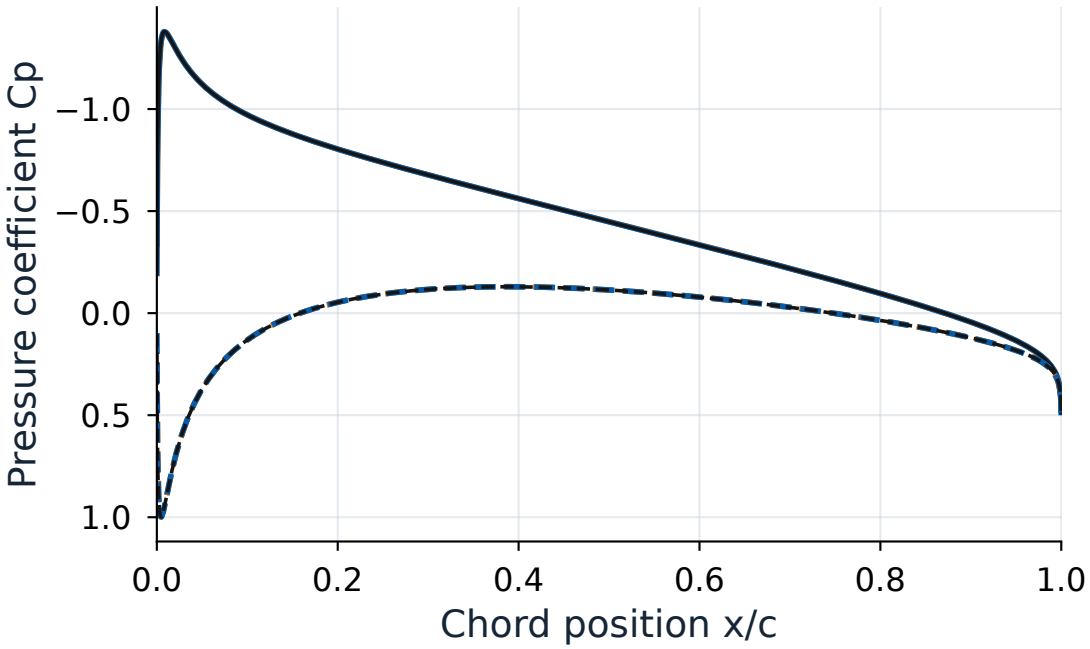
Joukowski airfoil: sharp cusp
Full chord



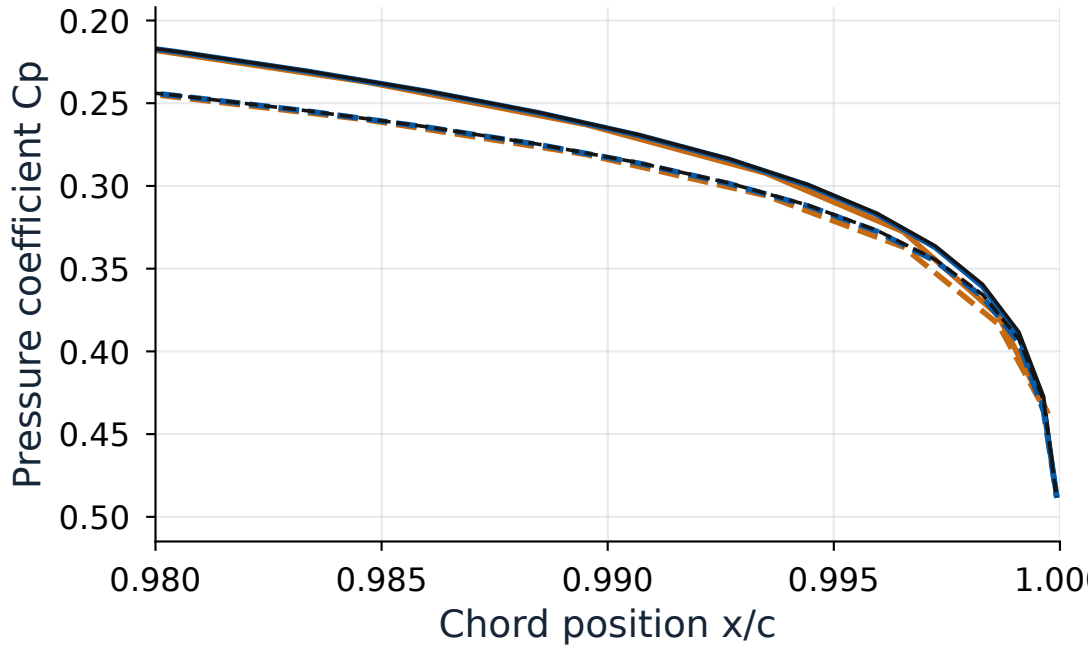
Joukowski airfoil: sharp cusp
Trailing-edge detail



Karman-Trefftz airfoil: 12-degree edge
Full chord



Karman-Trefftz airfoil: 12-degree edge
Trailing-edge detail



Negative Cp (suction) is upward. Right panels magnify the last 2% of chord.
All computed samples are retained; pressure is neither clipped nor smoothed.